

BETH LINDENBAUM

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EDUCATION

Duke University – Pratt School of Engineering

August 2023 – May 2027

Double Major - B.S.E. Electrical & Computer Engineering and Computer Science | GPA: 3.73/4.0

Durham, NC

Relevant Coursework: Nanoelectronic Devices, Computer Architecture, Machine Learning, Digital Systems, Microelectronic Devices & Circuits, Algorithm Design & Analysis, Data Structures & Algorithms, Signals & Systems, Fields & Waves, Data Bases, Linear Algebra, Calculus, Probability, Differential Equations, Chemistry, Physics, Biology

Extracurriculars: Women's Club Basketball Executive Committee, Duke Jewish Student Union Executive Committee, Duke Technology Scholars (DTech), Society of Women Engineers, Jewish Service Club Vice President

Awards: 2023 & 2024 Kevin Gorter Sport Club Memorial Scholarship, DTech Scholar

Clearview Regional High School

September 2019 – June 2023

Valedictorian – Rank 1/364 | GPA: 101.30/100

Mullica Hill, NJ

WORK EXPERIENCE

Undergraduate Nanoelectronics Researcher

January 2026 – Present

Franklin Group Lab - Duke University

Durham, NC

- Develop a sacrificial material-assisted lift-off printing process to fabricate 1- μ m channel length transistors with near-100% yield, overcoming resolution and scalability limits of conventional printed electronics.
- Demonstrate high-resolution additive transistor patterning using Hummink capillary flow printing and aerosol jet printing, enabling micron-scale feature definition without spin-coating or etching and significantly reducing chemical waste.

Embedded Linux Software Engineer Intern

June 2025 – August 2025

Schneider Electric

Raleigh, NC

- Develop a fully automated testing device using an ESP32 microcontroller to simulate BLE client behavior, Zigbee relay communication, and Wi-Fi messaging for Schneider's Energy Monitor system
- Program in C to test interactions between the Energy Monitor, iOS app, and home circuit breakers; collaborate with firmware written in Python and Rust to validate multi-protocol communication reliability

Teaching Assistant – Computer Architecture (COMPSCI 250)

January 2025 – May 2025

Duke University

Durham, NC

- Led weekly recitation sections and office hours. Provided individualized instruction to students, clarifying complex concepts and troubleshooting technical challenges. Collaborated with the professor and other TA's to ensure consistency in curriculum delivery
- Graded and assessed student projects and exams in a timely manner, providing valuable feedback for student development

PROJECTS

Pac-Man

Spring 2025

- Built a Pac-Man arcade game on FPGA using Verilog, MIPS assembly and a custom processor; displayed on VGA and joystick-controlled
- Integrated Verilog modules and assembly code via register-mapped I/O to track scoring, lives, and game state resets
- Designed FSMs for sprite movement, collision detection, animations, a high score system, and game reset logic (win/loss conditions)

Audio-visual Notification System for Drone Delivery

Fall 2023 – Spring 2025

- Designed, programmed and assembled an automatically activated and deactivated device, that simultaneously flashes multicolored LED lights and outputs a range of tones with Arduino IDE and microcontrollers
- Led a presentation of our product at Rural PRO-CARE Health Equity Research Network Annual Meeting 2024

LEADERSHIP EXPERIENCE

Duke Jewish Student Union Executive Committee – Communications Chair

Spring 2024 – Spring 2025

- Manage community outreach to the 700+ Jewish students at Duke University through email and social media
- Document weekly meeting minutes and distribute action items. Design graphics for Duke Jewish Student Union social media and merch

Duke Women's Club Basketball Executive Committee – Social Chair

Spring 2024 – Spring 2025

- Manage social calendar and plan, host, and monitor events for 55+ club members. Promote relationships with other campus organizations

TECHNICAL SKILLS

Languages: Java, Python, C, C++, Verilog, Assembly, MIPS, SQL, Swift, SwiftUI, HTML, CSS, JavaScript, Rust, LaTeX, Visual Basic

Technologies: FPGA, Microcontrollers, Arduino, MATLAB, Simulink, Linux, MacOS, Git, Bluetooth, BLE, Zigbee, Wifi, Microsoft Office

Other Skills: Firmware, Machine Learning, Scikit-learn, Jupyter Notebooks, Hummink, Aerosol Jet Printing, NumPy, Pandas, Soldering

CERTIFICATIONS

- Andrew Ng Supervised Machine Learning: Regression and Classification Course – DeepLearning.AI/Stanford Online